

ORIGINAL ARTICLE

Isolated Diastolic Hypertension and Cardiovascular Disease Risk in Young Adults

Rui Tang¹, Jaejin An¹, Andrew E. Moran¹, Daichi Shimbo¹, Norrina B. Allen¹, Vanessa Xanthakis¹, Kristi Reynolds¹, Yiyi Zhang¹

BACKGROUND: Whether isolated diastolic hypertension (IDH) in young adults represents a benign, age-specific phenotype or confers increased cardiovascular disease risk remains uncertain. We assessed the associations between hypertension subtypes, IDH, isolated systolic hypertension, and systolic-diastolic hypertension, with atherosclerotic cardiovascular disease and heart failure. We also characterized longitudinal transitions in hypertension subtypes from young adulthood into later life.

METHODS: Adults aged ≥ 18 years without cardiovascular disease were included from 4 US cohorts. Hypertension subtypes were defined using blood pressure thresholds of $\geq 130/80$ mm Hg. Cox proportional hazards models were used to estimate the associations of hypertension subtypes with atherosclerotic cardiovascular disease and heart failure.

RESULTS: Among 23957 participants (mean age, 44.5 years, 42.5% were young adults aged 18–39 years, 45.1% were male), 2320 atherosclerotic cardiovascular disease and 1243 heart failure events occurred over a median follow-up of 17.2 years. Among young adults, the adjusted hazard ratios (95% CI) for atherosclerotic cardiovascular disease were 1.36 (95% CI, 1.11–1.68) for IDH, 1.54 (95% CI, 1.02–2.32) for isolated systolic hypertension, and 1.84 (95% CI, 1.46–2.32) for systolic-diastolic hypertension, compared with normotension. Corresponding hazard ratios for heart failure were 1.69 (95% CI, 1.24–2.31) for IDH, 1.92 (95% CI, 1.09–3.40) for isolated systolic hypertension, and 1.69 (95% CI, 1.17–2.43) for systolic-diastolic hypertension. Associations were consistent among adults aged ≥ 40 years. Among young adults with IDH, 29.8% remained with IDH, whereas 36.5% progressed to systolic-diastolic hypertension later in life.

CONCLUSIONS: IDH in young adulthood represents an early marker of increased cardiovascular disease risk and a precursor to more adverse hypertension subtypes. These findings underscore the importance of early recognition, ongoing blood pressure monitoring, and guideline-concordant pharmacological treatment, when indicated, among young adults with IDH to reduce long-term cardiovascular disease risk. (*Hypertension*. 2026;83:2888–2898. DOI: 10.1161/HYPERTENSIONAHA.126.27333.) • **Supplement Material.**

Key Words: blood pressure ■ cardiovascular diseases ■ heart failure ■ isolated systolic hypertension ■ young adults

Isolated diastolic hypertension (IDH), defined as diastolic blood pressure (DBP) in the hypertensive range with systolic blood pressure (SBP) below the hypertensive threshold, is more common among younger adults and less prevalent in middle-aged or older adults.^{1–3} However, it remains uncertain whether IDH represents a benign, age-related phenomenon in young adults or confers an increased risk of cardiovascular disease (CVD). Previous studies have reported inconsistent associations between IDH and CVD outcomes. Analyses of

US cohorts and the UK Biobank found no association between IDH and CVD risk,^{4,5} whereas analyses of the Japan Medical Data Center database reported a higher risk of CVD among individuals with IDH.⁶ These discrepancies may be partly attributable to differences in the DBP threshold used to define IDH (DBP ≥ 80 versus ≥ 90 mm Hg), the choice of reference group (normotension based on both SBP and DBP versus non-IDH), and differences in study populations (younger versus older adults). Furthermore, analyses of the Korean National

Correspondence to: Yiyi Zhang, PhD, Division of General Medicine, Columbia University Irving Medical Center, 622 W 168th St, PH9-103J, New York, NY 10032. Email yz3160@cumc.columbia.edu

Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/HYPERTENSIONAHA.126.27333>.

© 2026 American Heart Association, Inc.

Hypertension is available at www.ahajournals.org/journal/hyp

NOVELTY AND RELEVANCE

What Is New?

In this pooled analysis of 4 US cohorts, isolated diastolic hypertension in young adults aged 18 to 39 years was associated with higher risks of atherosclerotic cardiovascular disease and heart failure compared with normotension. More than one-third of young adults with isolated diastolic hypertension progressed to systolic-diastolic hypertension later in life.

What Is Relevant?

Isolated diastolic hypertension in young adulthood is not a benign phenotype, but rather an early marker of elevated cardiovascular disease risk and a precursor to more adverse hypertension subtypes.

Clinical/Pathophysiological Implications?

Early recognition of isolated diastolic hypertension in young adults may help identify individuals at elevated long-term cardiovascular risk. Lifestyle-based risk reduction, longitudinal blood pressure monitoring, and guideline-concordant pharmacological treatment, when indicated, may help reduce long-term cardiovascular risk in this population.

Nonstandard Abbreviations and Acronyms

ARIC	Atherosclerosis Risk in Communities
ASCVD	atherosclerotic cardiovascular disease
BP	blood pressure
CARDIA	Coronary Artery Risk Development in Young Adults
CVD	cardiovascular disease
DBP	diastolic blood pressure
FHS	Framingham Heart Study
HF	heart failure
HR	hazard ratio
IDACO	International Database on Ambulatory Blood Pressure in Relation to Cardiovascular Outcomes
IDH	isolated diastolic hypertension
ISH	isolated systolic hypertension
JHS	Jackson Heart Study
MESA	Multi-Ethnic Study of Atherosclerosis
SBP	systolic blood pressure
SDH	systolic-diastolic hypertension

Health Insurance Service database suggested that IDH may be more strongly associated with CVD risk among young adults aged 20 to 39 years than among older age groups, with associations attenuated with advancing age.^{7,8} Whether IDH confers a consistent CVD risk among US young adults remains unclear.

Young adults with IDH may also experience transitions over time to systolic-diastolic hypertension (SDH: both SBP and DBP in the hypertensive range) or other hypertension subtypes later in life, but the prevalence and longitudinal patterns of these transitions remain

poorly characterized in diverse US cohorts. Moreover, the optimal DBP level for minimizing CVD risk among young adults with IDH remains unknown.^{9–12} The 2025 American Heart Association/American College of Cardiology blood pressure (BP) guideline has identified this evidence gap as a critical area for future research.¹

The current study sought to: (1) assess the associations of IDH with incident atherosclerotic CVD (ASCVD) and heart failure (HF) among young adults in the United States; (2) characterize within-person longitudinal transitions in hypertension subtypes from young adulthood into later life; and (3) identify DBP levels associated with the lowest risk of ASCVD and HF among individuals with SBP <130 mm Hg.

METHODS

Data Availability Statement

Data used in this study may be requested from the Biologic Specimen and Data Repository Information Coordinating Center repository and the database of Genotypes and Phenotypes. In addition, investigators with an approved manuscript proposal or ancillary study proposal may request data directly from the study coordinating centers: BioLINCC: <https://biolincc.nhlbi.nih.gov/>; dbGaP: <https://www.ncbi.nlm.nih.gov/gap/>; CARDIA (Coronary Artery Risk Development in Young Adults): <https://sites.uab.edu/cardia/>; FHS (Framingham Heart Study; including data from the Offspring, Third Generation, Omni 1, and Omni 2 cohorts): <https://www.framinghamheartstudy.org/>; JHS (Jackson Heart Study): <https://www.jacksonheartstudy.org/Research/Study-Data/Data-Access>; MESA (Multi-Ethnic Study of Atherosclerosis): <https://mesa-nhlbi.org/>.

Study Design and Cohorts

Because of the sensitive nature of the data collected for this study and restrictions from participating cohort studies,

qualified researchers trained in human subject confidentiality protocols may request access through the respective cohort study coordinating centers, in accordance with established data access procedures for each cohort study.

This study analyzed data from 4 large, population-based prospective epidemiological cohorts in the United States: (1) CARDIA¹³; (2) FHS^{14–16}; (3) JHS¹⁷; and (4) MESA.¹⁸ The designs of each cohort have been described previously and are detailed in the [Supplemental Methods](#). All study protocols were approved by institutional review boards at participating institutions, and all participants provided written informed consent.

In the current analysis, we included participants aged ≥ 18 years ([Figure S1](#)). We excluded individuals with a history of CVD at baseline ($n=756$), defined as a history of acute coronary syndrome, myocardial infarction, stroke, transient ischemic attack, HF, cardiac arrest, stable or unstable angina, coronary revascularization procedures, or peripheral arterial disease. We also excluded individuals with missing data on hypertension subtype ($n=1294$), traditional CVD risk factors required for multivariable adjustment ($n=733$), or follow-up for incident ASCVD and HF events ($n=619$). The final analytic sample included 23 957 participants. The first visit at which participants met all inclusion and exclusion criteria was designated as the baseline for this analysis. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology reporting guidelines.

Clinical Data Collection

Demographic, traditional ASCVD and HF risk factors, and BP measurements were assessed using standardized protocols in each study.^{13–18} Information on age, sex, race, ethnicity, smoking status, history of CVD, and medication use was self-reported. Laboratory measurements included fasting glucose, low-density lipoprotein cholesterol, and high-density lipoprotein cholesterol. Diabetes was defined as a fasting glucose level ≥ 126 mg/dL, use of antidiabetic medication, or self-reported diabetes status.

BP Measurement and Subtype Classification

BP was measured by trained staff using standardized methods in each cohort. In CARDIA, BP was measured 3 $\times$ at each study visit, and the average of the second and third readings was used for the analysis. From year 0 to 15, BP was measured using a random-zero sphygmomanometer (Hawksley & Sons, Ltd); beginning at year 20, BP was measured with an automated oscillometric device (Omron HEM907XL).¹⁹ The oscillometric values were calibrated to the random-zero scale using study-specific equations to ensure comparability across examinations.²⁰ In FHS, BP was measured twice at each visit using a mercury sphygmomanometer, and the average of the 2 readings was used.²¹ In JHS, BP was measured twice, and the average of the 2 readings was used. At visit 1, BP was measured using a random-zero sphygmomanometer (Hawksley & Sons, Ltd); at visits 2 and 3, BP was measured using an oscillometric device (Omron HEM-907XL). Random-zero values were calibrated to the oscillometric device using robust regression.²² In MESA, BP was measured 3 $\times$ using an automated oscillometric device (Dinamap PRO 100). The average of the second and third measurements was used for analysis.²³

In all 4 cohorts, we classified individuals into 4 hypertension subtypes using their baseline BP: (1) normotension, defined as SBP <120 mmHg and DBP <80 mmHg without antihypertensive medication use; (2) IDH, defined as SBP <130 mmHg and DBP ≥ 80 mmHg; (3) isolated systolic hypertension (ISH), defined as SBP ≥ 130 mmHg and DBP <80 mmHg; and (4) SDH, defined as SBP ≥ 130 mmHg and DBP ≥ 80 mmHg.¹ Individuals with SBP <120 mmHg and DBP <80 mmHg who were taking antihypertensive medication were not classified as normotensive, because their untreated BP phenotype could not be reliably determined. These individuals, along with those with SBP 120 to 129 mmHg and DBP <80 mmHg, were not assigned to any of the 4 prespecified BP phenotype categories and were excluded from the analyses.

Follow-Up and Outcome Events

The outcomes were time to the first incident ASCVD event and time to the first incident HF event. ASCVD was defined as nonfatal myocardial infarction, death from coronary heart disease, and fatal or nonfatal stroke. Events were ascertained and adjudicated using the specific protocol of each cohort.^{13–18} Additional details are provided in the [Supplemental Methods](#).

Statistical Analysis

Participant characteristics at the baseline visit were summarized for the overall pooled cohort, by individual study, and by age group (18–39 and ≥ 40 years). Incidence rates of ASCVD and HF events were calculated overall, by hypertension subtype, and within age group.

To examine the associations between hypertension subtypes at baseline and the risk of incident ASCVD and HF events, we used Cox proportional hazards regression with age as the time scale after confirming that the assumption of proportional hazards was met. In primary analysis, we performed stratified analyses by age (18–39 versus ≥ 40 years), given our a priori hypothesis that the association between hypertension subtypes and CVD may differ by age. In secondary analyses, we also examined heterogeneity by sex (male versus female), race and ethnicity (White versus non-White), body mass index (<30 versus ≥ 30 kg/m²), diabetes status, and smoking status (noncurrent versus current), as these are key demographic and clinical factors relevant to cardiovascular risk. Statistical interactions were assessed using likelihood ratio tests. Models were adjusted for established cardiovascular risk factors. Model 1 was adjusted for sex, race, and ethnicity. Model 2 was further adjusted for body mass index, smoking status, low-density lipoprotein cholesterol, high-density lipoprotein cholesterol, antihypertensive medication use, lipid-lowering therapy, and diabetes status. To account for potential cohort effects, all models were stratified by study cohort, allowing the baseline hazard function to vary across cohorts.

To characterize longitudinal transitions in hypertension subtypes during follow-up, we assessed a subgroup of young adult participants who had follow-up BP measurements available during the middle-to-older adulthood period (50–69 years). For each participant, the hypertension subtype was defined based on BP measurements from the earliest available visit within each period. Pairwise transitions between subtypes were summarized and visualized with a Sankey diagram.

To determine age-specific (18–39 versus ≥40 years) DBP levels associated with the lowest risk of ASCVD and HF among adults with SBP <130 mmHg, we modeled DBP as a continuous variable using restricted cubic splines in Cox proportional hazards models. Nonlinearity was assessed by testing the significance of the nonlinear spline terms.

For sensitivity analyses, we restricted the analyses to participants not taking antihypertensive medication therapy at baseline and censored follow-up at the initiation of therapy. In addition, we redefined hypertension subtypes using the alternative BP thresholds of 140/90 mmHg.²⁴ All analyses were conducted using R version 4.5.0 (R Foundation for Statistical Computing). Two-sided *P*<0.05 was considered statistically significant.

RESULTS

A total of 23957 participants were included in the analyses, with a mean (SD) age of 44.5 (16.9) years. Among the participants, 10 191 (42.5%) were young adults aged 18 to 39 years, 10805 (45.1%) were male, and 13535 (56.5%) self-identified as non-Hispanic White (Table). At baseline, 12 371 (51.6%) participants had normotension, 3669 (15.3%) had IDH, 3534 (14.8%) had ISH, and 4383 (18.3%) had SDH. Detailed characteristics of the study participants by individual study are shown in Table S1.

During a median follow-up of 17.2 years, a total of 2320 incident ASCVD and 1243 incident HF events

Table. Characteristics of Participants by Age*

Characteristic	Overall (N=23957)	Age, 18–39 y (n=10191)	Age, ≥40 y (n=13766)
Age, y	44.5±16.9	28.3±5.7	56.6±11.4
Study			
CARDIA	5070 (21.2%)	5060 (49.7%)	10 (0.1%)
FHS	9444 (39.4%)	4728 (46.4%)	4716 (34.3%)
JHS	3108 (13.0%)	403 (4.0%)	2705 (19.6%)
MESA	6335 (26.4%)	0	6335 (46.0%)
Sex			
Male	10805 (45.1%)	4635 (45.5%)	6170 (44.8%)
Female	13152 (54.9%)	5556 (54.5%)	7596 (55.2%)
Race and ethnicity			
Non-Hispanic White	13535 (56.5%)	7010 (68.8%)	6525 (47.4%)
Non-Hispanic Black	7685 (32.1%)	3037 (29.8%)	4648 (33.8%)
Other	2737 (11.4%)	144 (1.4%)	2593 (18.8%)
BMI, kg/m ²	27.0±5.9	25.1±5.4	28.5±5.9
Smoking status			
Not smoking	12775 (53.3%)	5450 (53.5%)	7325 (53.2%)
Previously smoked	5753 (24.0%)	1577 (15.5%)	4176 (30.3%)
Currently Smoking	5429 (22.7%)	3164 (31.0%)	2265 (16.5%)
SBP, mmHg	121.4±19.0	112.5±12.5	128.0±20.3
DBP, mmHg	74.3±10.8	71.9±10.5	76.1±10.7
Hypertension subtypes			
Normotension	12371 (51.6%)	7341 (72.0%)	5030 (36.5%)
IDH	3669 (15.3%)	1685 (16.5%)	1984 (14.4%)
ISH	3534 (14.8%)	271 (2.7%)	3263 (23.7%)
SDH	4383 (18.3%)	894 (8.8%)	3489 (25.3%)
Total cholesterol, mg/dL	191.3±37.2	181.3±34.7	198.7±37.4
LDL-C, mg/dL	118.1±34.4	112.1±32.8	122.6±34.9
HDL-C, mg/dL	52.7±15.1	52.7±14.0	52.7±15.8
Diabetes	1577 (6.6%)	100 (1.0%)	1477 (10.7%)
Antihypertensive medication	3973 (16.6%)	141 (1.4%)	3832 (27.8%)
Lipid-lowering medication	1840 (7.7%)	50 (0.5%)	1790 (13.0%)

BMI indicates body mass index; CARDIA, Coronary Artery Risk Development in Young Adults; DBP, diastolic blood pressure; FHS, Framingham Heart Study; HDL-C, high-density lipoprotein cholesterol; IDH, isolated diastolic hypertension; ISH, isolated systolic hypertension; JHS, Jackson Heart Study; LDL-C, low-density lipoprotein cholesterol; MESA, Multi-Ethnic Study of Atherosclerosis; SBP, systolic blood pressure; and SDH, systolic and diastolic hypertension.
*Numbers are mean (SD) for normally distributed data and n (%) for categorical variables.

occurred (Table S2). Among young adults aged 18 to 39 years, the hazard ratios (HRs) for ASCVD were 1.36 (95% CI, 1.11–1.68) for IDH, 1.54 (95% CI, 1.02–2.32) for ISH, and 1.84 (95% CI, 1.46–2.32) for SDH in the multivariable-adjusted model, compared with normotension (Figure 1). Similarly, the HRs for HF were 1.69 (95% CI, 1.24–2.31) for IDH, 1.92 (95% CI, 1.09–3.40) for ISH, and 1.69 (95% CI, 1.17–2.43) for SDH, compared with young adults with normotension (Figure 2). No significant interactions between hypertension subtypes and age were observed for ASCVD or HF ($P_{\text{interaction}}=0.87$ and 0.38, respectively). In secondary analyses, no significant interactions were observed for ASCVD across any subgroup factor (all $P_{\text{interaction}} > 0.05$; Figure S2). For HF,

significant interactions were observed for race and ethnicity and diabetes status (both $P_{\text{interaction}}=0.04$).

In sensitivity analyses excluding participants taking antihypertensive medication at baseline and censoring follow-up at medication initiation ($n=19984$), associations between hypertension subtypes with ASCVD and HF remained similar to those observed in the main analyses (Figures S3 and S4). In addition, redefining hypertension subtypes using an alternative BP threshold of 140/90 mm Hg found consistent associations (Table S3; Figures S5 and S6).

Among 5459 young adults with follow-up BP measurements available during middle-to-older adulthood (50–69 years), the prevalence of normotension, IDH,

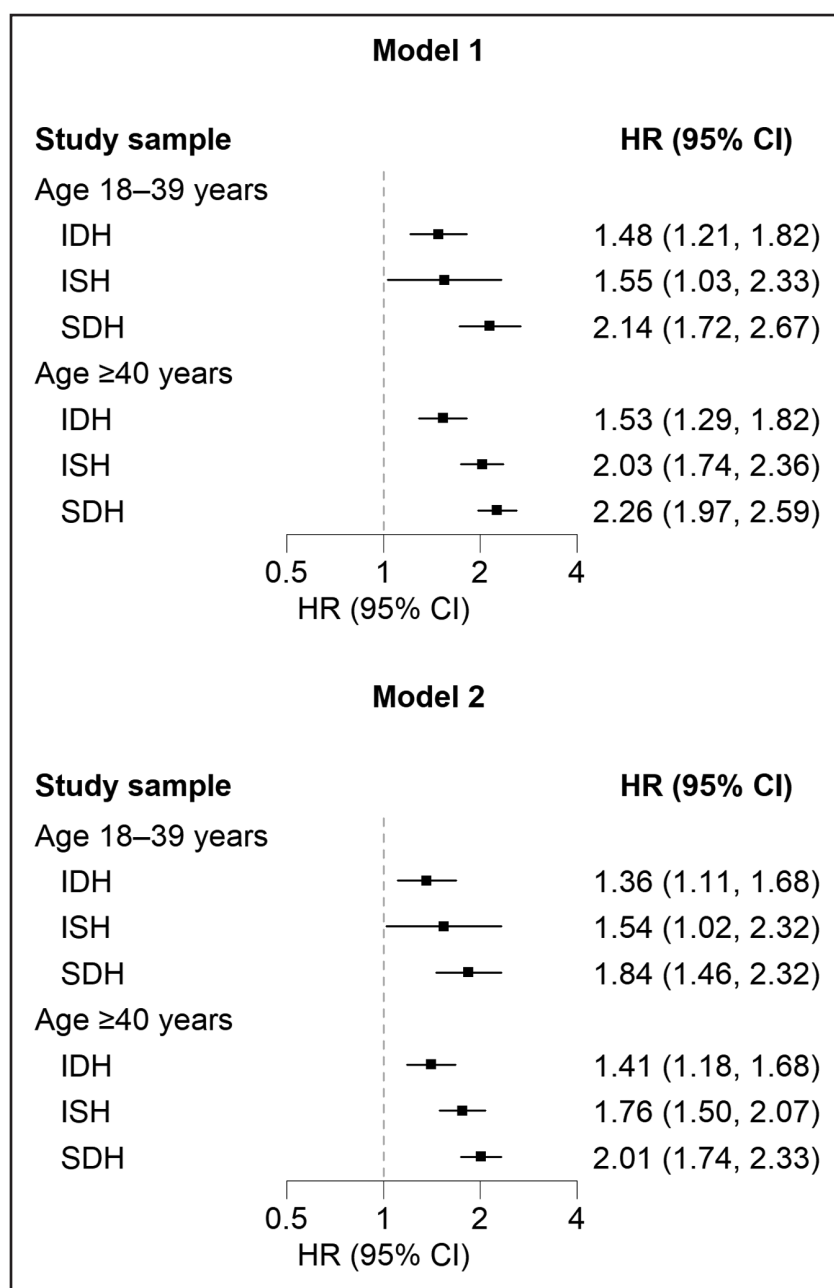


Figure 1. Associations between hypertension subtypes and atherosclerotic cardiovascular disease (ASCVD).

Model 1: adjusted for sex, race, and ethnicity. Model 2: model 1+body mass index, smoking status, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, diabetes, antihypertensive medication, and lipid-lowering medication. All models were stratified by the study cohort. HR indicates hazard ratio; IDH, isolated diastolic hypertension; ISH, isolated systolic hypertension; and SDH, systolic-diastolic hypertension.

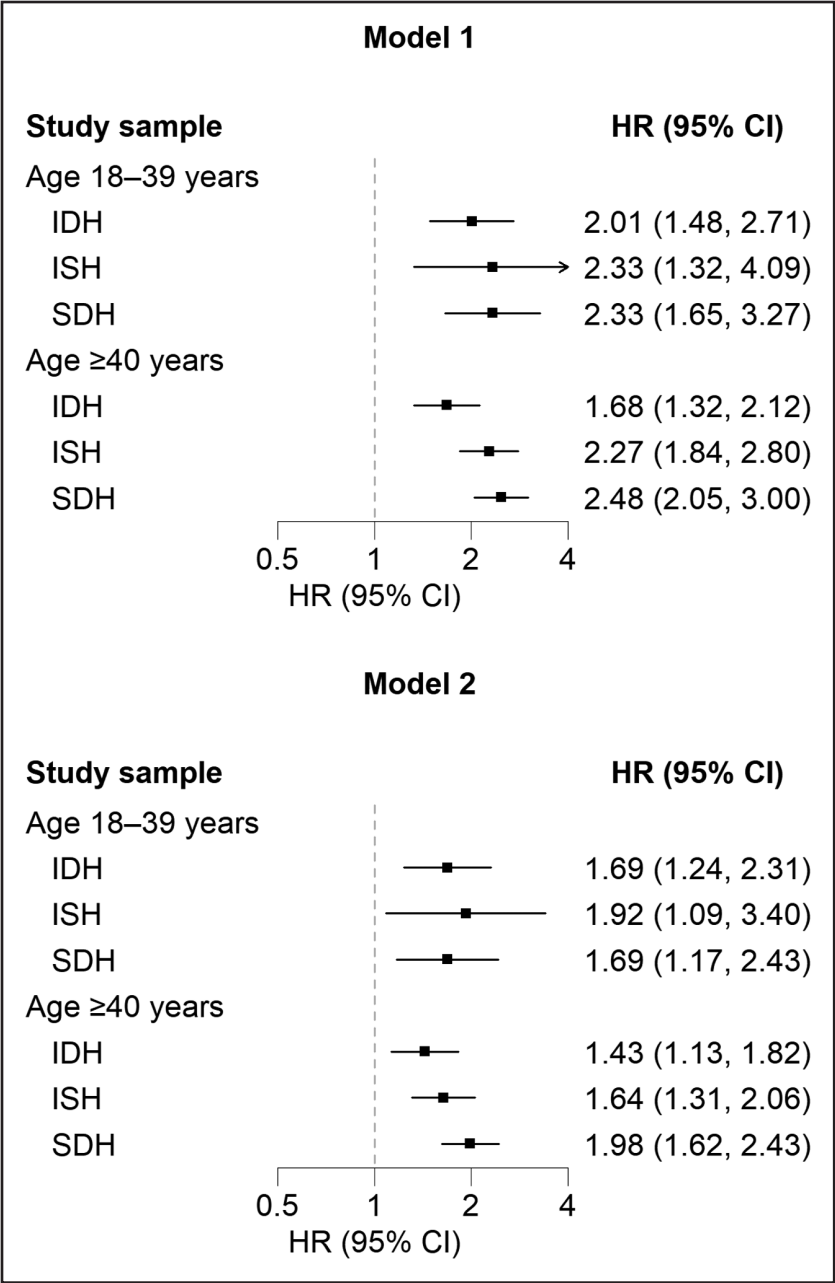


Figure 2. Associations between hypertension subtypes and heart failure (HF).
Model 1: adjusted for sex, race, and ethnicity. Model 2: model 1+body mass index, smoking status, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, diabetes, antihypertensive medication, and lipid-lowering medication. All models were stratified by the study cohort. HR indicates hazard ratio; IDH, isolated diastolic hypertension; ISH, isolated systolic hypertension; and SDH, systolic-diastolic hypertension.

ISH, and SDH was 70.8%, 17.0%, 2.4%, and 9.7%, respectively, in young adulthood; the corresponding prevalence in middle-to-older adulthood was 46.7%, 18.0%, 8.0%, and 27.3%, respectively (Figure 3). Among young adults with IDH, 29.8% remained with IDH, 24.3% transitioned to normotension, 9.4% to ISH, and 36.5% to SDH during middle-to-older adulthood. Compared with young adults with IDH who remained with IDH in later life, those who transitioned to SDH were more likely to be female and non-Hispanic Black and had higher levels of SBP and high-density lipoprotein cholesterol (Table S4). Sensitivity analyses excluding participants who were taking antihypertensive medications at baseline and censoring follow-up at medication initiation (n=5409)

showed similar transition patterns. Among young adults with IDH, 29.7% remained with IDH, 25.0% transitioned to normotension, 9.2% to ISH, and 36.1% to SDH during middle-to-older adulthood (Figure S7). Among young adults with SBP <130 mmHg, restricted cubic spline models suggested a nonlinear association between DBP and ASCVD ($P_{\text{nonlinearity}}=0.006$), with lower DBP associated with a lower risk of ASCVD at least down to a DBP of 66 mmHg after adjustment for CVD risk factors and SBP (Figure 4). A similar nonlinear association was observed for HF among adults aged ≥40 years ($P_{\text{nonlinearity}}=0.002$), whereas the association between DBP and HF appeared to be linear among young adults ($P_{\text{nonlinearity}}=0.18$; Figure 5).

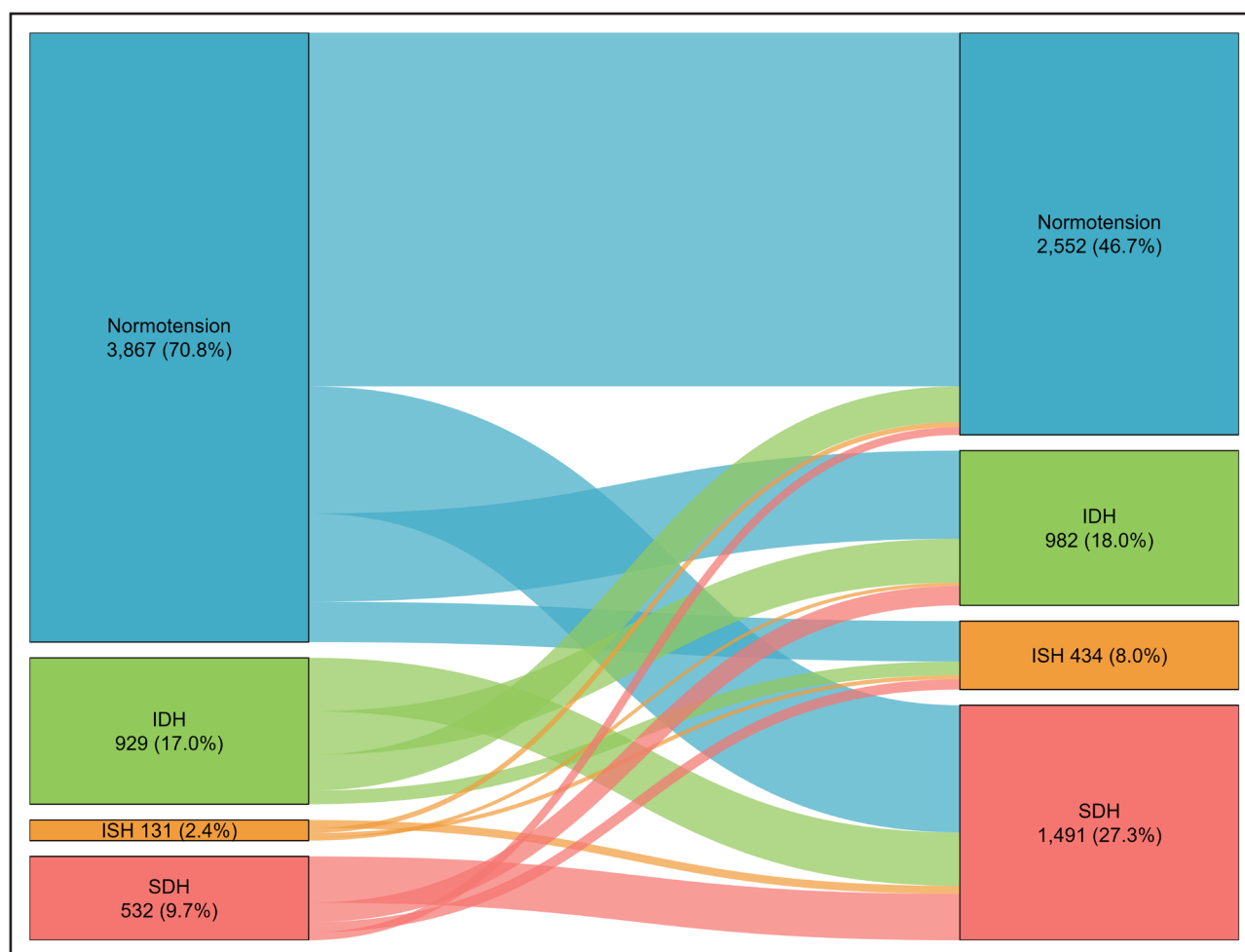


Figure 3. Sankey plot of hypertension subtype transitions from young adulthood (18–39 years) to middle-to-older adulthood (50–69 years).

IDH indicates isolated diastolic hypertension; ISH, isolated systolic hypertension; and SDH, systolic-diastolic hypertension.

DISCUSSION

In this analysis of 23 957 adults from 4 large US prospective cohort studies, the prevalence of IDH was 16.5% among young adults aged 18 to 39 years and 14.4% among those aged ≥ 40 years. Young adults with IDH, ISH, or SDH had significantly higher risks of ASCVD and HF compared with those with normotension, and these associations were consistent across age groups (18–39 versus ≥ 40 years). Longitudinally, progression to SDH was the most common trajectory among young adults with IDH at baseline, of whom 36.5% transitioned to SDH later in life. Among young adults with SBP < 130 mmHg, DBP showed a nonlinear association with ASCVD, with lower DBP associated with a lower risk of ASCVD at least down to a DBP of 66 mmHg, whereas the relationship with HF appeared to be linear. These findings suggest that IDH in young adulthood represents an early marker of elevated CVD risk and a precursor to more adverse hypertension subtypes.

Prior studies of IDH have yielded conflicting results, with some suggesting that it is a benign phenotype and

others associating it with increased CVD risk. In analyses of middle-aged and older adults, IDH generally showed no significant association with incident CVD. For example, in the ARIC study (Atherosclerosis Risk in Communities; $n=8703$; mean age, 56 years), IDH was not associated with incident ASCVD (HR, 1.06 [95% CI, 0.89–1.26]) or HF (HR, 0.91 [95% CI, 0.76–1.09]) compared with no IDH (SBP < 130 mmHg and DBP < 80 mmHg).⁴ Similarly, in MESA ($n=5104$; mean age, 60 years), IDH was not associated with incident CVD (HR, 1.19 [95% CI, 0.77–1.84]) compared with no IDH.²⁵ In the UK Biobank ($n=151\,831$; mean age, 54 years), IDH was not associated with CVD compared with no IDH when defined by BP thresholds of 130/80 mmHg, but showed a modest association (HR, 1.15 [95% CI, 1.04–1.29]) when defined using the higher threshold (SBP < 140 mmHg and DBP ≥ 90 mmHg versus SBP < 140 mmHg and DBP < 90 mmHg).⁵ In contrast, analyses of young adults aged 20 to 39 years from the Korean National Health Insurance Service ($N=6\,424\,090$) found higher CVD risk with IDH compared with normotension,⁷ and in an expanded cohort ($N=8\,109\,484$; ages, 20–89 years), risk associations

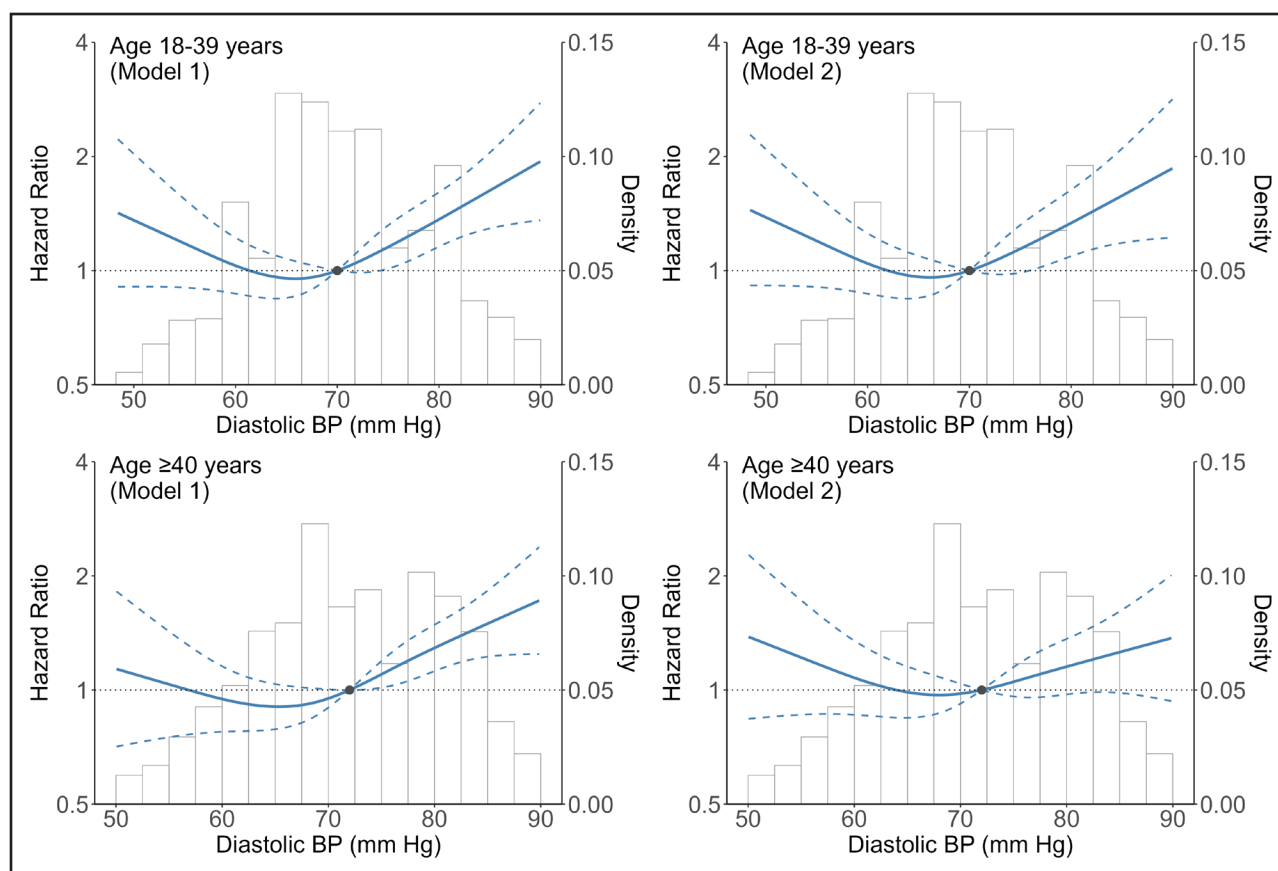


Figure 4. Restricted cubic spline modeling the relationship between diastolic blood pressure (DBP) and atherosclerotic cardiovascular disease (ASCVD) among individuals with systolic blood pressure (SBP) <130 mm Hg, stratified by age.

Model 1: adjusted for sex, race, and ethnicity, body mass index, smoking status, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, diabetes, antihypertensive medication, and lipid-lowering medication. Model 2: model 1+SBP. All models were stratified by the study cohort. Reference point (black dot) was set at the median DBP.

were modified by age and stronger at younger ages.⁸ In addition, the IDACO study (International Database on Ambulatory Blood Pressure in Relation to Cardiovascular Outcomes) reported that IDH based on 24-hour ambulatory BP monitoring was associated with higher cardiovascular risk among adults aged <50 years (HR, 2.87 [95% CI, 1.72–4.80]) but not among older adults.²⁶ The current analysis extends prior evidence by focusing specifically on a diverse cohort of US young adults and found that IDH was associated with higher risks of ASCVD and HF compared with normotension. Importantly, the magnitude of these associations was similar across age groups (18–39 versus ≥40 years), suggesting that the adverse cardiovascular implications of IDH may begin early in adulthood and persist over time. These findings were robust across multiple sensitivity analyses, including redefining hypertension subtypes using alternative BP thresholds, suggesting that prior inconsistent results in the literature may reflect differences in study design and population characteristics rather than the absence of cardiovascular risk associated with IDH.

These findings are clinically relevant in the context of evolving hypertension guidelines. The 2025 American

Heart Association/American College of Cardiology BP guideline classifies DBP of 80 to 89 mm Hg as stage 1 hypertension and recommends consideration of antihypertensive therapy when BP remains ≥130/80 mm Hg after an initial 3 to 6 months of lifestyle modification to prevent target organ damage and mitigate further BP progression.¹ In contrast, the 2024 European Society of Cardiology hypertension guideline defines IDH using a higher threshold of SBP <140 mm Hg and DBP ≥90 mm Hg and recognizes IDH as a phenotype associated with modestly increased relative cardiovascular risk, particularly among younger adults.²⁷ However, the European Society of Cardiology guideline notes that the absolute cardiovascular risk among individuals with IDH is generally low and that it remains unclear whether IDH should routinely prompt initiation of BP-lowering medication, while emphasizing the need for follow-up because individuals with IDH are at increased risk for developing systolic hypertension.²⁷ Findings from the current study support the view that IDH in young adulthood should not be dismissed as benign, while underscoring the need for future studies to determine whether early pharmacological treatment, beyond

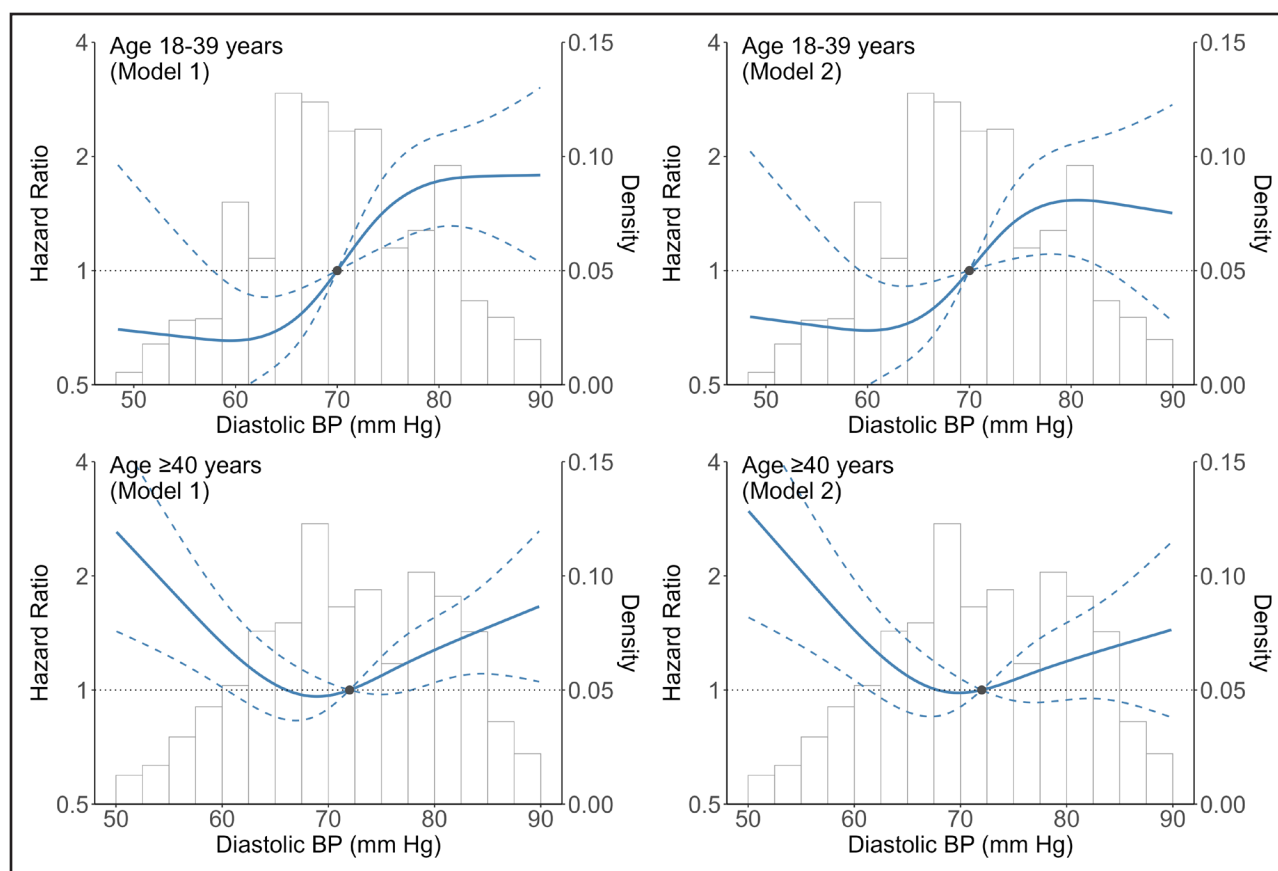


Figure 5. Restricted cubic spline modeling the relationship between diastolic blood pressure (DBP) and heart failure (HF) among individuals with systolic blood pressure (SBP) <130 mm Hg, stratified by age.

Model 1: adjusted for sex, race, and ethnicity, body mass index, smoking status, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, diabetes, antihypertensive medication, and lipid-lowering medication. Model 2: model 1+SBP. All models were stratified by the study cohort. Reference point (black dot) was set at the median DBP.

lifestyle intervention and monitoring, reduces long-term cardiovascular risk.

Hypertension subtypes are dynamic and evolve over the life course. In the current study, 29.8% of young adults with IDH remained with IDH in later life, whereas 36.5% progressed to SDH. Few prior studies have characterized such long-term trajectories of hypertension subtypes. In FHS, IDH was more likely to progress to SDH than to ISH, and ISH or SDH rarely reverted to IDH.²⁸ In MESA, baseline IDH strongly predicted the development of ISH or SDH in later life.²⁵ These transitions are biologically plausible, as arterial stiffening with age raises SBP while DBP plateaus or falls, resulting in widened pulse pressure.^{29,30} Taken together, these findings suggest that IDH in young adulthood is neither benign nor stable, but rather represents an early marker of elevated CVD risk and subsequent progression to more harmful hypertension subtypes such as SDH. Clinically, this underscores the importance of early recognition and ongoing monitoring of young adults with IDH, as timely intervention may help prevent progression to SDH and mitigate long-term CVD risk.^{31,32} Although most BP-lowering trials have focused on SBP targets,

the optimal DBP target, particularly among young adults, remains unclear.^{9–12} A post hoc analysis pooling data from the Systolic Blood Pressure Intervention Trial and the Action to Control Cardiovascular Risk in Diabetes BP trial found that DBP levels of 70 to 80 mmHg were associated with the lowest CVD risk among individuals with treated SBP <130 mmHg, with higher CVD risk observed at both lower and higher DBP values.³³ Similarly, in the Treating to New Targets trial of patients with coronary artery disease, DBP showed a J-shaped relationship with future cardiovascular events, with the lowest risk observed at a DBP of 81 mmHg.³⁴ The current analysis extends these observations from high-risk older adults to lower-risk young adults and found a nonlinear association between DBP and ASCVD when SBP was <130 mmHg, with lower DBP associated with a lower risk of ASCVD at least down to a DBP of 66 to 68 mmHg in both young and older adults. In contrast, the pattern for HF differed by age: among middle-aged and older adults, the spline analyses suggested that the lower DBP was associated with a lower risk of HF at least down to a DBP of around 70 mmHg, whereas among young adults, the association appeared approximately

linear without clear evidence of a nonlinear nadir. This difference may reflect age-related vascular changes, such as increased arterial stiffness and impaired diastolic function, which make older adults more vulnerable to inadequate coronary perfusion at lower DBP levels, whereas younger adults may tolerate lower DBP without the same hemodynamic compromise.^{29,35,36} Alternatively, the relatively small number of HF events among young adults may have limited our ability to detect nonlinear associations in this group.

Strengths and Limitations

This study has several strengths, including the pooling of data from 4 large prospective cohorts with high-quality exposure and outcome assessments, large sample sizes of both young and older adults, and long follow-up durations. This study also has several limitations. First, heterogeneity in BP measurement protocols across cohorts and reliance on single-visit BP measurements for hypertension subtype assignment may have led to misclassification. Second, most young adults in this analysis were from CARDIA and FHS, whereas participants aged ≥ 40 years were primarily from FHS, JHS, and MESA. This raises the possibility of cohort effects in addition to age effects. Although we stratified Cox models by cohort to allow baseline hazards to vary, residual cohort-related influences cannot be fully excluded. Finally, the current study population comprised primarily non-Hispanic White and non-Hispanic Black adults, with relatively fewer participants from Hispanic, Asian, or other racial and ethnic groups. This may limit the generalizability of our findings to those underrepresented populations.

Perspective

In this analysis of 23957 adults from 4 large US cohorts, IDH, ISH, and SDH were each associated with higher risks of ASCVD and HF compared with normotension, with similar associations across age groups. More than one-third of young adults with IDH progressed to SDH later in life. These findings suggest that IDH is not a benign phenotype in young adults and underscore the importance of early recognition, lifestyle-based risk reduction, ongoing BP monitoring, and guideline-concordant pharmacological treatment when indicated among young adults with IDH to reduce long-term CVD risk.

ARTICLE INFORMATION

Received May 04, 2026; accepted July 31, 2026.

Affiliations

Division of General Medicine (R.T., A.E.M., Y.Z.) and Division of Cardiology (D.S.), Columbia University Irving Medical Center, New York, NY. Department of Research & Evaluation, Kaiser Permanente Southern California, Pasadena (J.A.,

K.R.). Department of Health Systems Science, Kaiser Permanente Bernard J. Tyson School of Medicine, Pasadena, CA (J.A., K.R.). Departments of Preventive Medicine and Medicine (Cardiology), Northwestern University, Chicago, IL (N.B.A.). Section of Preventive Medicine and Epidemiology, Department of Medicine, Boston University, MA (V.X.).

Acknowledgments

The authors thank all the investigators, staff, and participants of the CARDIA study (Coronary Artery Risk Development in Young Adults), the FHS (Framingham Heart Study), the JHS (Jackson Heart Study), and the MESA (Multi-Ethnic Study of Atherosclerosis) for their valuable contributions. This manuscript was reviewed and approved by the Publications and Presentations Committee of each study.

Sources of Funding

This work was supported by the National Institutes of Health (NIH) grants R01HL155081, R01HL168379, and R01HL158790 (Y. Zhang and J. An). The funding organizations had no role in the design and conduct of the study; the collection, management, analysis, and interpretation of the data; the preparation, review, or approval of the manuscript; and the decision to submit the manuscript for publication. The views expressed in this manuscript are those of the authors and do not necessarily represent the views of the National Heart, Lung, and Blood Institute (NHLBI), the NIH, or the US Department of Health and Human Services. The CARDIA study (Coronary Artery Risk Development in Young Adults) is supported by contracts 75N92023D00002, 75N92023D00003, 75N92023D00004, 75N92023D00005, and 75N92023D00006 from the NHLBI. The FHS (Framingham Heart Study) was supported by NHLBI (HHSN2682015000011 and 75N92025D00012) and the Boston University School of Medicine. The JHS (Jackson Heart Study) is supported by contracts HHSN268201800010I, HHSN268201800011I, HHSN268201800012I, HHSN268201800013I, HHSN268201800014I, and HHSN268201800015I from the NHLBI, with additional support from the National Institute on Minority Health and Health Disparities (NIMHD). The MESA (Multi-Ethnic Study of Atherosclerosis) was supported by contracts 75N92025D00022, 75N92020D00001, HHSN268201500003I, N01-HC-95159, 75N92025D00026, 75N92020D00005, N01-HC-95160, 75N92020D00002, N01-HC-95161, 75N92025D00024, 75N92020D00003, N01-HC-95162, 75N92025D00027, 75N92020D00006, N01-HC-95163, 75N92025D00025, 75N92020D00004, N01-HC-95164, 75N92025D00028, 75N92020D00007, N01-HC-95165, N01-HC-95166, N01-HC-95167, N01-HC-95168, and N01-HC-95169 from the NHLBI, and by grants UL1-TR-000040, UL1-TR-001079, and UL1-TR-001420 from the National Center for Advancing Translational Sciences (NCATS).

Disclosures

None.

Supplemental Material

Supplemental Methods
Tables S1–S4
Figures S1–S7
References 13–18

REFERENCES

1. Jones DW, Ferdinand KC, Taler SJ, Johnson HM, Shimbo D, Abdalla M, Altieri MM, Bansal N, Bello NA, Bress AP, et al; Writing Committee Members. 2025 AHA/ACC/AANP/AAPA/ABC/ACCP/ACPM/AGS/AMA/ASPC/NMA/PCNA/SGIM guideline for the prevention, detection, evaluation and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2025;152:e114–e218. doi: 10.1161/CIR.0000000000001356
2. Xie K, Gao X, Bao L, Shan Y, Shi H, Li Y. The different risk factors for isolated diastolic hypertension and isolated systolic hypertension: a national survey. *BMC Public Health*. 2021;21:1672. doi: 10.1186/s12889-021-11686-9
3. Franklin SS, Jacobs MJ, Wong ND, L'Italien GJ, Lapuerta P. Predominance of isolated systolic hypertension among middle-aged and elderly US hypertensives: analysis based on National Health and Nutrition Examination Survey (NHANES) III. *Hypertension*. 2001;37:869–874. doi: 10.1161/01.hyp.37.3.869
4. McEvoy JW, Daya N, Rahman F, Hoogveen RC, Blumenthal RS, Shah AM, Ballantyne CM, Coresh J, Selvin E. Association of isolated diastolic hypertension as defined by the 2017 ACC/AHA blood pressure guideline

with incident cardiovascular outcomes. *JAMA*. 2020;323:329–338. doi: 10.1001/jama.2019.21402

5. McGrath BP, Kundu P, Daya N, Coresh J, Selvin E, McEvoy JW, Chatterjee N. Isolated diastolic hypertension in the UK Biobank: comparison of ACC/AHA and ESC/NICE Guideline definitions. *Hypertension*. 2020;76:699–706. doi: 10.1161/HYPERTENSIONAHA.120.15286
6. Kaneko H, Itoh H, Yotsumoto H, Kiriyaama H, Kamon T, Fujiu K, Morita K, Michihata N, Jo T, Takeda N, et al. Association of isolated diastolic hypertension based on the cutoff value in the 2017 American College of Cardiology/American Heart Association Blood Pressure Guidelines with subsequent cardiovascular events in the general population. *J Am Heart Assoc*. 2020;9:e017963. doi: 10.1161/JAHA.120.017963
7. Lee H, Yano Y, Cho SMJ, Park JH, Park S, Lloyd-Jones DM, Kim HC. Cardiovascular risk of isolated systolic or diastolic hypertension in young adults. *Circulation*. 2020;141:1778–1786. doi: 10.1161/CIRCULATIONAHA.119.044838
8. Lee H, Yano Y, Cho SMJ, Park S, Lloyd-Jones DM, Kim HC. Cardiovascular risk of isolated diastolic hypertension defined by the 2017 American College of Cardiology/American Heart Association Blood Pressure Guideline: a nationwide age-stratified cohort study. *Hypertension*. 2020;76:e44–e46. doi: 10.1161/HYPERTENSIONAHA.120.16018
9. SPRINT Research Group. A randomized trial of intensive versus standard blood-pressure control. *N Engl J Med*. 2015;373:2103–2116. doi: 10.1056/NEJMoa1511939
10. ACCORD Study Group. Effects of intensive blood-pressure control in type 2 diabetes mellitus. *N Engl J Med*. 2010;362:1575–1585. doi: 10.1056/NEJMoa1001286
11. Liu J, Li Y, Ge J, Yan X, Zhang H, Zheng X, Lu J, Li Xi, Gao Y, Lei L, et al; ESPRIT Collaborative Group. Lowering systolic blood pressure to less than 120 mm Hg versus less than 140 mm Hg in patients with high cardiovascular risk with and without diabetes or previous stroke: an open-label, blinded-outcome, randomised trial. *Lancet*. 2024;404:245–255. doi: 10.1016/S0140-6736(24)01028-6
12. Zhang W, Zhang S, Deng Y, Wu S, Ren J, Sun G, Yang J, Jiang Y, Xu X, Wang TD, et al; STEP Study Group. Trial of intensive blood-pressure control in older patients with hypertension. *N Engl J Med*. 2021;385:1268–1279. doi: 10.1056/NEJMoa2111437
13. Friedman GD, Cutter GR, Donahue RP, Hughes GH, Hulley SB, Jacobs DR, Liu K, Savage RJ. CARDIA: study design, recruitment, and some characteristics of the examined subjects. *J Clin Epidemiol*. 1988;41:1105–1116. doi: 10.1016/0895-4356(88)90080-7
14. Feinleib M, Kannel WB, Garrison RJ, McNamara PM, Castelli WP. The Framingham Offspring Study. Design and preliminary data. *Prev Med*. 1975;4:518–525. doi: 10.1016/0091-7435(75)90037-7
15. Splansky GL, Corey D, Yang Q, Atwood LD, Cupples LA, Benjamin EJ, D'Agostino RB, Fox CS, Larson MG, Murabito JM, et al. The Third Generation Cohort of the National Heart, Lung, and Blood Institute's Framingham Heart Study: design, recruitment, and initial examination. *Am J Epidemiol*. 2007;165:1328–1335. doi: 10.1093/aje/kwm021
16. Tsao CW, Vasan RS. Cohort profile: the Framingham Heart Study (FHS): overview of milestones in cardiovascular epidemiology. *Int J Epidemiol*. 2015;44:1800–1813. doi: 10.1093/ije/dyv337
17. Sempos CT, Bild DE, Manolio TA. Overview of the Jackson Heart Study: a study of cardiovascular diseases in African American men and women. *Am J Med Sci*. 1999;317:142–146. doi: 10.1097/00000441-199903000-00002
18. Bild DE, Bluemke DA, Burke GL, Detrano R, Diez Roux AV, Folsom AR, Greenland P, Jacob DR, Kronmal R, Liu K, et al. Multi-ethnic study of atherosclerosis: objectives and design. *Am J Epidemiol*. 2002;156:871–881. doi: 10.1093/aje/kwf113
19. Yano Y, Ning H, Allen N, Reis JP, Launer LJ, Liu K, Yaffe K, Greenland P, Lloyd-Jones DM. Long-term blood pressure variability throughout young adulthood and cognitive function in midlife: the Coronary Artery Risk Development in Young Adults (CARDIA) study. *Hypertension*. 2014;64:983–988. doi: 10.1161/HYPERTENSIONAHA.114.03978
20. Jacobs DR Jr, Yatsuya H, Hearst MO, Thyagarajan B, Kalhan R, Rosenberg S, Smith LJ, Barr RG, Duprez DA. Rate of decline of forced vital capacity predicts future arterial hypertension: the Coronary Artery Risk Development in Young Adults Study. *Hypertension*. 2012;59:219–225. doi: 10.1161/HYPERTENSIONAHA.111.184101
21. Vasan RS, Larson MG, Leip EP, Kannel WB, Levy D. Assessment of frequency of progression to hypertension in non-hypertensive participants in the Framingham Heart Study: a cohort study. *Lancet*. 2001;358:1682–1686. doi: 10.1016/S0140-6736(01)06710-1
22. Seals SR, Colantonio LD, Tingle JV, Shimbo D, Correa A, Griswold ME, Muntner P. Calibration of blood pressure measurements in the Jackson Heart Study. *Blood Press Monit*. 2019;24:130–136. doi: 10.1097/MBP.0000000000000379
23. Shimbo D, Shea S, McClelland RL, Viera AJ, Mann D, Newman J, Lima J, Polak JF, Psaty BM, Muntner P. Associations of aortic distensibility and arterial elasticity with long-term visit-to-visit blood pressure variability: the Multi-Ethnic Study of Atherosclerosis (MESA). *Am J Hypertens*. 2013;26:896–902. doi: 10.1093/ajh/hpt040
24. Chobanian AV, Bakris GL, Black HR, Cushman WC, Green LA, Izzo JL, Jones DW, Materson BJ, Oparil S, Wright JT, et al; Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure. National Heart, Lung, and Blood Institute. Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure. *Hypertension*. 2003;42:1206–1252. doi: 10.1161/01.HYP.0000107251.49515.c2
25. Jacobsen AP, Al Rifai M, Arps K, Whelton SP, Budoff MJ, Nasir K, Blaha MJ, Psaty BM, Blumenthal RS, Post WS, et al. A cohort study and meta-analysis of isolated diastolic hypertension: searching for a threshold to guide treatment. *Eur Heart J*. 2021;42:2119–2129. doi: 10.1093/eurheartj/ehab111
26. McEvoy JW, Yang WY, Thijs L, Zhang ZY, Melgarejo JD, Boggia J, Hansen TW, Asayama K, Ohkubo T, Dolan E, et al; International Database on Ambulatory Blood Pressure in Relation to Cardiovascular Outcomes (IDACO) Investigators. Isolated diastolic hypertension in the IDACO study: an age-stratified analysis using 24-hour ambulatory blood pressure measurements. *Hypertension*. 2021;78:1222–1231. doi: 10.1161/HYPERTENSIONAHA.121.17766
27. McEvoy JW, McCarthy CP, Bruno RM, Brouwers S, Canavan MD, Ceconi C, Christodorescu RM, Daskalopoulou SS, Ferro CJ, Gerdts E, et al; ESC Scientific Document Group. 2024 ESC Guidelines for the management of elevated blood pressure and hypertension. *Eur Heart J*. 2024;45:3912–4018. doi: 10.1093/eurheartj/ehae178
28. Bourdillon MT, Song RJ, Musa Yola I, Xanthakis V, Vasan RS. Prevalence, predictors, progression, and prognosis of hypertension subtypes in the Framingham Heart Study. *J Am Heart Assoc*. 2022;11:e024202. doi: 10.1161/JAHA.121.024202
29. Vlachopoulos C, O'Rourke M. Diastolic pressure, systolic pressure, or pulse pressure? *Curr Hypertens Rep*. 2000;2:271–279. doi: 10.1007/s11906-000-0010-6
30. Yano Y, Neeland IJ, Ayers C, Peshock R, Berry JD, Lloyd-Jones DM, Greenland P, Mitchell GF, Vongpatanasin W. Hemodynamic and mechanical properties of the proximal aorta in young and middle-aged adults with isolated systolic hypertension: the Dallas Heart Study. *Hypertension*. 2017;70:158–165. doi: 10.1161/HYPERTENSIONAHA.117.09279
31. Stamler J. Epidemiologic findings on body mass and blood pressure in adults. *Ann Epidemiol*. 1991;1:347–362. doi: 10.1016/1047-2797(91)90045-e
32. Franklin SS, Pio JR, Wong ND, Larson MG, Leip EP, Vasan RS, Levy D. Predictors of new-onset diastolic and systolic hypertension: the Framingham Heart Study. *Circulation*. 2005;111:1121–1127. doi: 10.1161/01.CIR.0000157159.39889.EC
33. Li J, Somers VK, Gao X, Chen Z, Ju J, Lin Q, Mohamed EA, Karim S, Xu H, Zhang L. Evaluation of optimal diastolic blood pressure range among adults with treated systolic blood pressure less than 130 mm Hg. *JAMA Netw Open*. 2021;4:e2037554. doi: 10.1001/jamanetworkopen.2020.37554
34. Bangalore S, Messerli FH, Wun CC, Zuckerman AL, DeMicco D, Kostis JB, LaRosa JC; Treating to New Targets Steering Committee and Investigators. J-curve revisited: an analysis of blood pressure and cardiovascular events in the Treating to New Targets (TNT) Trial. *Eur Heart J*. 2010;31:2897–2908. doi: 10.1093/eurheartj/ehq328
35. Boutouyrie P, Chowienczyk P, Humphrey JD, Mitchell GF. Arterial stiffness and cardiovascular risk in hypertension. *Circ Res*. 2021;128:864–886. doi: 10.1161/CIRCRESAHA.121.318061
36. Chrysant SG, Chrysant GS. The age-related hemodynamic changes of blood pressure and their impact on the incidence of cardiovascular disease and stroke: new evidence. *J Clin Hypertens*. 2013;16:87–90. doi: 10.1111/jch.12253